Measuring Extreme Weather and its Effects on Agricultural Productivity in the United States

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Abstract

Extreme weather events, such as heatwaves, cold frosts, droughts, floods, and extreme precipitation, can significantly influence agricultural productivity. However, when raw weather data are interpolated across space, localized extremes may be attenuated which can influence the estimated impact of extreme weather on agricultural outcomes. To address this, this study compares measures calculated directly from weather station data and then interpolated ("pre-transformed measures") with those derived after interpolating raw weather data ("post-transformed measures"). It then evaluates their effects on corn yields across the Corn Belt Region in the United States using fixed-effects regression models. The results show notable differences in the distribution of extreme weather variables under pre- and post-interpolation. Although the estimated effects of some measures, such as cooling degree days (CDD) and killing degree days (KDD), are relatively similar across methods, others (e.g., growing degree days (GDD) and extreme wet conditions) show statistically significant differences. Counterfactual analyses shows that yield predictions based on post-interpolated data tend to be systematically higher, reflecting the method's tendency to understate localized extremes. A Shapley Decomposition analysis identifies heat stress as the predominant driver of the observed differences in yield predictions. The findings highlight the need to preserve local weather extremes and variability in agricultural impact analyses, emphasizing that more refined methods are crucial for accurately capturing the effects of extreme weather on agricultural production.

Keywords: extreme weather, measurement error, production
JEL codes: O13, Q15, Q54

1 Introduction

Extreme weather events, from severe droughts and heatwaves to intense precipitation and floods, pose significant challenges to global agriculture. These events can significantly affect crop yields and disrupt food supply chains, incurring huge economic losses. Historical events such as the 1930s Dust Bowl in the United States show how devastating the consequences of extreme weather can be for agricultural production. More recently, the 2012 drought led to substantial declines in U.S. crop

yields and farm incomes. These events are neither confined to the United States nor exceptional: with rising global temperatures, many regions now face an increased frequency and intensity of weather extremes (IPCC, 2023a).

For example, climate science assessments affirm with high confidence that more frequent extreme weather will reduce food production (Porter et al., 2014). Historically, the 1930s Dust Bowl drought severely affected U.S. farmlands. More recently, the nationwide 2012 drought, the worst in over half a century caused sharp declines in crop yields. These events highlight the vulnerability of the agricultural sector to weather extremes and the growing risks associated with climate change. As global temperatures rise, the frequency and intensity of these weather extremes are expected to increase (IPCC, 2023b), increasing concerns about global food security. This trend highlights growing worries about how vulnerable agriculture is and the wider consequences for global food systems.

At the heart of understanding and examining the impact of extreme weather on agriculture lies a fundamental measurement challenge. Weather is inherently variable across space and time; capturing the intensity and spatial heterogeneity of extreme weather at fine scales is difficult. Conventional gridded climate datasets and ground-based weather stations often sacrifice local precision for broader coverage, effectively smoothing out the sharp peaks and troughs that define true extremes (Haylock et al., 2008). This smoothing can mask the true magnitude and frequency of events like heat spikes or extreme precipitation, creating biases estimates in downstream analyses that rely on these data. In precipitation data, for example, localized extremes may be underrepresented if the nearest station misses the heaviest rainfall. For temperature, an intense heatwave can become diluted through spatial averaging, leading to underestimation of extreme heat days.

Such measurement challenges have direct consequences for economic and policy analyses. A rich body of research demonstrates how extreme weather disrupts crop yields, farm revenues, and productivity growth (Schlenker & Roberts, 2009; Lobell et al., 2011; Burke & Emerick, 2016). However, many of these findings rely on weather data that have already undergone interpolation and post-processing, potentially obscuring localized extremes and local variability. From an econometric standpoint, measurement error in climate variables tends to bias estimated coefficients in

yield-response models (Lobell, 2013). In classical measurement-error settings, the regression coefficients are often attenuated, understating true impacts. Worse still, if errors correlate with other explanatory variables or timing mismatches (Li & Ortiz-Bobea, 2022), more complex biases can arise. Studies have shown that even moderate noise in precipitation data can lead researchers to underestimate or mischaracterize the magnitude of yield losses by a factor of two or more (Lobell, 2013). Similarly, two-way fixed effects estimators, commonly used in panel data settings, can further amplify these biases if the within-group variation in weather data has been "smoothed away".

A key methodological question is how best to capture, and then model extreme weather. While many empirical studies compute non-linear transformations of climate data (e.g., Growing Degree Days, Cooling Degree Days, Killing Degree Days) after spatial interpolation, it is important to recognize that the order of operations might matter greatly. As a simple example, mathematically, $(\frac{x+y}{2})^2$ is not the same as $\frac{x^2+y^2}{2}$; the former inherently smooths outliers, leading to an underrepresentation of extremes. When extreme weather metrics are calculated from already-interpolated data, the smoothing that occurs during interpolation can systematically understate the true intensity of extremes. Consequently, the estimated impacts of these events on crop outcomes may have attenuation bias which can potentially distort policy decisions, for instance, undercutting the perceived need for heat-tolerant seed varieties, additional irrigation infrastructure, or more robust insurance programs. In addition to attenuation bias, the smoothing of outliers can reduce within variation when working with panel data. As a consequence, panel estimators using within variation to identify the effects of extreme weather can lead to estimated impacts that are much larger in magnitude relative to their true effects. The logic is simple—when extremes are attenuated, within variation is reduced, thus we are attributing within changes in the outcome variable to artificially small within changes in our explanatory variable of interest. The consequence is an inflation of the anticipated effects of climate change.

Given the immense economic and food security stakes, it is crucial to improve how we measure and analyze extreme weather in agricultural contexts. Policymakers rely on these estimates to allocate resources, design disaster relief, and guide agricultural resilience strategies. In the United States alone, crop insurance programs have paid out over $118 billion for weather-related losses

from 2001–2022, underscoring the financial severity of the problem. As extreme weather intensifies, these costs will likely rise, highlighting the importance of accurate measurement of these events.

This paper takes an interdisciplinary perspective, integrating insights from climatology, geospatial science, and econometrics, to assess how different approaches to measuring extreme weather shape estimates of agricultural impacts. By comparing methods that compute key weather indicators before versus after spatial interpolation, it aims to pinpoint the magnitude and direction of measurement error arising from smoothing. These findings of the paper contribute to the growing literature on climate and agriculture by offering insights to improve the accuracy and reliability of extreme weather metrics.

The remainder of this paper is structured as follows. Section 2 reviews the broader literature on extreme weather impacts in agriculture. Section 3 describes the data sources and methodological framework, detailing the rationale for exploring pre- and post-interpolation measures of weather extremes. Section 4 presents results, demonstrating how these different approaches yield systematically different estimates for extreme weather variables and, in turn, different yield impact coefficients. Section 5 discusses implications for policy and future research. Finally, Section 6 concludes by summarizing key findings and outlining areas for future research.

2 Literature Review

Climate change has led to an increase in the frequency and intensity of extreme weather events, which pose significant challenges for agriculture. The latest assessment by the IPCC (2022) highlights that these changes are already impacting crop production. For example, the report shows that between 1983 and 2009 around three-quarters of global harvested area experienced yield losses due to drought, with estimated losses of $166 billion. Researchers widely agree that rising temperatures and shifting precipitation patterns driven by climate change have mostly negative effects on crop yields in vulnerable regions. Each step of the food supply chain, from on-farm production to storage and transport, is increasingly strained by climate extremes. Building on these observations, a substantial body of research has since explored how extreme weather affects agricultural

production and how farmers and systems respond.

Early econometric studies leveraged year-to-year random weather fluctuations to isolate climate impacts on agriculture. A study by Deschênes & Greenstone (2007) used U.S. county-level panel data on weather and farm profits to estimate the impacts of climate change. Their surprising result was that moderate warming might increase annual agricultural profits by about 4%. In other words, they did not find the large damages one might expect; in fact, their point estimates suggested a modest benefit of warmer temperatures. This counter-intuitive finding spurred debate. The authors themselves cautioned that the traditional "hedonic" approach (linking climate to land values) yielded highly unstable estimates, sensitive to modeling choices. By focusing on short-run weather variability, their study arguably captured farmers' immediate responses (like coping mechanisms within a year) but did not fully account for extreme events or long-run adjustments. Critics noted that extreme heat occurrences were relatively infrequent in their sample, and subsequent research identified some errors affecting their results (Fisher et al., 2012). Thus, while the study was influential in showing the value of panel approaches, its optimistic conclusions were viewed as limited by methodology.

In contrast to early findings of minimal damage, a body of later research found much more pessimistic outcomes when accounting for extreme weather. Schlenker & Roberts (2009) introduced a nonlinear approach to examine U.S. crop yield responses to temperature. They discovered that crop yields increase with temperature up to a critical threshold (about 29°C for corn, 30°C for soybeans), but temperatures beyond those thresholds cause yields to decline. In fact, days of extreme heat above these optimum points are "very harmful" to crop growth, leading to significant yield losses. Notably, the damage from excessive heat was far greater than the benefit of warmer days below the threshold, indicating an asymmetric impact. Furthermore, by comparing results from both year-to-year fluctuations and cross-sectional spatial differences, Schlenker & Roberts (2009) found a similar nonlinear pattern, which suggests limited historical adaptation to extreme heat. In other words, even farmers in historically hotter regions (who presumably have adapted their practices to a warmer climate) suffer large yield declines when temperatures exceed the same critical levels. This lack of observed adaptation implies that if extreme heat days become more common,

current farming practices and crop varieties may be ill-prepared to cope. These findings sharply contrast with Deschênes and Greenstone's results and underscore that accounting for extreme tail events is crucial – failing to do so can mask the true risks to agriculture. Consistent with this, other studies focusing on disasters show substantial impacts. The literature thus shifted toward recognizing that extreme weather events (as opposed to moderate changes in averages) drive the most significant economic losses in agriculture.

Beyond purely statistical analyses of historical data, researchers have also employed crop simulation models and cross-disciplinary assessments to project future impacts and explore adaptation options. Rosenzweig et al. (2014) coordinated an inter-comparison of global crop models, finding that under high-emission scenarios, major staples are projected to suffer severe yield declines, especially in tropical regions, absent significant adaptation measures. Similarly, a meta-analysis by Challinor et al. (2014) of dozens of studies found that unmitigated climate change is likely to reduce yields of key crops (like wheat, rice, and corn) in both temperate and tropical areas, particularly from mid-century onward, and that adaptive measures (e.g. changing planting dates, crop varieties, or expanding irrigation) can partially offset some losses. These modeling studies complement econometric findings by allowing controlled "what-if" experiments, such as simulating crop outcomes with and without adaptation or with future CO_2 fertilization effects. The general convergence is that extremes will pose increasingly severe risks: for instance, frequency of extreme agricultural drought could double or more in many regions at 2°C warming and above, and yield variability is expected to rise. However, these models come with their own limitations. Many crop models simplify or exclude how farmers change behavior in response to climate (beyond simple adaptations), and they may not capture socio-economic factors like market adjustments or policy responses.

Additionally, projections are only as good as their inputs: uncertainties in climate models (especially regarding rainfall extremes or storm frequency) translate into uncertainties in agricultural impacts. Therefore, while there is broad agreement that the net effect of climate change on agriculture is negative globally (especially for poorer regions), the magnitude varies across studies due to different assumptions about adaptation, CO_2 fertilization, technology improvements, and how extremes are represented. A critical observation from the literature is that short-run and long-run

effects may differ – farmers can often accommodate modest weather changes in the short term, but persistent shifts and more frequent extremes may overwhelm current adaptation capacity.

As the literature has evolved, recognition has grown that final results hinge critically on measuring extreme weather with accuracy. Many recent papers rely on gridded climate datasets (such as satellite-derived or reanalysis products) to measure rainfall, temperature, and extreme events across study areas. However, measurement errors in weather data can bias impact estimates if not properly addressed. In particular, if the error in the weather variables is systematic (i.e. correlated with the true weather conditions or with geography) rather than purely random, it is termed a non-classical measurement error. Unlike classical random error (which on average just adds noise and typically biases estimates toward zero), non-classical errors have a bias structure that can lead to over- or under-estimation of weather impacts in unknown ways (Adams et al., 2021). For example, if a rainfall product consistently overestimates precipitation amounts during heavy storms, an analysis might wrongly conclude that extreme rainfall has little effect on yields (because the data never shows truly extreme values when in reality crops might have been damaged by floods). In contrast, an opposite bias (underestimating rainfall) could exaggerate the apparent sensitivity of crops to drought. In other words, the pattern of errors matters. Studies have documented that widely used weather datasets are not error-free and often deviate in systematic ways. An evaluation of modern precipitation datasets over Asia, for instance, found that they all tended to overcount light rains and miss a portion of high-intensity rainfall event. This kind of bias means researchers must be cautious: the data errors are not just random that averages out, but can mislead conclusions if, say, drought impacts are analyzed with a dataset that smooths out extreme dryness. Similarly, one reanalysis dataset (ERA5) has been noted to over-estimate rainfall relative to ground observations in some regions, and to under-report the most extreme downpours (Xin et al., 2022). Such biases align with the notion of non-classical measurement error, since the error depends on the actual value (e.g. heavier rains have larger underestimation).

Temporal aggregation is another source of error. Many earlier studies aggregated weather data annually or seasonally, potentially "smoothing away" brief but damaging heatwaves or deluges. Although widespread availability of daily data has enabled more precise definitions of extreme

events (e.g., degree days, days above 35°C, or prolonged dry spells), it also raises the stakes for data fidelity: a day or two of misplaced extreme temperatures can affect estimated yield sensitivities. Misalignment of growing seasons, as noted by Li & Ortiz-Bobea (2022), can introduce additional non-classical measurement errors when the timing of planting and harvest does not match the fixed calendar period used in weather data.

Despite the recognized importance of accurate weather metrics, methodological gaps remain in how to systematically compare different data processing choices. While some research discusses spatial interpolation or threshold definitions, few studies examine how the sequence of data processing itself, i.e., whether extreme metrics are derived before or after interpolation—might influence the final estimates of climate impacts on agriculture.

Addressing this gap, the present study focuses on how two common approaches to capturing extreme weather can yield different results: (1) deriving extreme weather measures first ("pre-transformed measures"), then interpolating, versus (2) interpolating raw daily data first and then deriving extreme measures ("post-transformed measures"). Concentrating on the U.S. Corn Belt, this paper demonstrates that these methodological choices can potentially alter the estimated effects of extreme weather on agricultural outcomes. By focusing on this aspect of measurement error, this research aims to provide more accurate estimates of the impact of extreme weather on agriculture, which is essential for developing effective agricultural policies and strategies in the face of climate change.

3 Data and Methodology

3.1 Data Sources and Study Region

The primary weather data source for this study is the Global Historical Climatology Network-Daily (GHCN-D) dataset, provided by the National Oceanic and Atmospheric Administration (NOAA). The GHCN-D dataset contains daily observations of several climate variables, including maximum temperature, minimum temperature, precipitation and the elevation of the station.

This paper focuses on the Corn Belt region of the United States, which roughly spans ten major

corn-producing states: Iowa, Illinois, Indiana, Missouri, Minnesota, South Dakota, Nebraska, Ohio, Wisconsin, and Michigan. This region offers high agricultural productivity supported by a relatively dense network of weather stations. Critically, it also experiences temperature and precipitation extremes, allowing for robust estimates of non-linear temperature and precipitation effects.

Weather stations are selected based on how actively they recorded data during the study period (1950–2023) of this paper. Stations with more than 5% missing values are excluded to ensure sufficient data coverage, resulting in a final sample of 314 stations distributed across the region (Figure 1).

In addition to GHCN-D, elevation data from Amazon Web Services (AWS) Digital Elevation Model (DEM) is used to incorporate DEM-based elevation information to improve spatial interpolation accuracy, especially when temperature or precipitation patterns are strongly influenced by topography. Each grid cell is assigned an elevation value from the DEM and included in interpolation models.

County-level yield data for major crops, including corn, comes from the United States Department of Agriculture's National Agricultural Statistics Service (USDA-NASS). Since we focus specifically on corn productivity, we limit the analysis to counties and years with valid corn yield data.

3.2 Pre- vs. Post-Interpolation

A key methodological contribution of this paper is to compare two different sequences for deriving extreme weather variables. First, weather measures (e.g., Growing Degree Days, Killing Degree Days) are calculated from raw weather station data prior to spatial interpolation. These weather measures are then interpolated over space ($0.05° \times 0.05°$ gridded surface) using geospatial techniques (e.g., Inverse Distance Weighting (IDW)). Second, this process is reversed, i.e., the raw weather station data are first interpolated over the same space, followed by calculating the weather measures from the gridded data.

Because the weather effects on crop growth are generally non-linear, calculating metrics like GDD or KDD typically involves non-linear transformations of temperature. Jensen's Inequality

states the function of an expectation is not the same as the expectation of a function. Translated to the context of this paper, the non-linear transformation of an average (e.g., interpolated temperature) will differ systematically from the average of non-linear transformations (e.g., derived from raw station data).

For instance, if temperature on a given day at two stations is 25°C and 35°C, the post-interpolation method may average these observations to 30°C before computing the GDD or KDD, smoothing out the extreme. In contrast, the pre-interpolation approach first applies the non-linear GDD/KDD function at each station and only then aggregates or interpolates. Because GDD and KDD grow more rapidly above certain thresholds (e.g., 29°C), failing to capture localized extremes can lead to biased estimates—often underestimating the true impact of hot days. Jensen's Inequality thus provides a formal explanation of why these two approaches can yield systematically different estimates of extreme weather exposure.

3.3 Weather Measures

For weather measures, this study calculates several non-linear metrics (see appendix). Growing Degree Days (GDD), Killing Degree Days (KDD) and Cooling Degree Days are calculated using the sinusoidal function of the daily minimum and maximum temperatures and estimated by integrating the area under the sine curve for each day and summed over growing and planting seasons (Figures 2a-3). Following Annan & Schlenker (2015), the bounding temperatures for corn for the calculation of GDD are taken as 10-29°C, while the threshold for Killing Degree Days (extreme damaging heat) is taken as temperature above 29°C.

For precipitation extremes, we use the Standardized Precipitation Index (SPI) which fits a gamma distribution (McKee et al., 1993) using Maximum Likelihood Estimation to historical precipitation time series data to establish a relationship between precipitation values and their probability of occurrence. The gamma distribution is then transformed to normal distribution to standardize precipitation deviations from normal. The resulting deviation values are SPI values, where positive values indicate wet conditions and negative SPI values indicate drought conditions. Table 1 below shows the range of values for SPI and their category. In order to focus on the extremes only, we

adjust the SPI index by focusing only on values below -1.0 for drought and values greater than +1 for extreme wetness.

3.4 Spatial Aggregation to County-Level

To aggregate grid-level spatial data to county level, this paper uses the 2023 CDL (Cropland Data Layer) raster data to calculate the share of corn within each grid by overlaying the CDL raster data on the grid. Each grid is intersected with county boundaries to determine its spatial overlap with that county and weights are assigned based on the proportion of the area of the cell that lies within the county. These weights are then adjusted by the corn share from the CDL raster data to account for the distribution of the crop within each cell (Figure 4).

Aggregation to a county level is necessary because corn yield data from the USDA-NASS is only available at the county level. The use of corn-specific weights ensures that the aggregated weather variables are more representative of the areas where corn is grown within each county.

3.5 Econometric Specification

The study then estimates various specifications of the following relationship:

$$\log(\text{yield}_{ijt}) = \alpha_0 + (\beta \cdot X_{ijt}) + C_i + Y_t + \varepsilon_{ijt} \tag{1}$$

where $\log(\text{yield}_{ijt})$ is the logarithm of crop yield j of county i in year t. β includes weather and crop-specific yield effects of the different weather measures to be estimated. X_{ijt} represents the vector of different crop-specific extreme weather events. C_i denotes the county-level fixed effects, Y_t includes the year-fixed effect, and ε_{ijt} is the idiosyncratic error term.

The coefficient vector β is of primary interest, which tells gives us an estimate of how each extreme weather measure affects yield. Comparing β estimates from the pre-interpolation approach to those from the post-interpolation approach reveals how measurement error tied to interpolation can lead to systematically different estimates of weather impacts.

4 Results

4.1 Descriptive Statistics

Figures 5 and 6 compare (pre-interpolation vs. post-interpolation) values of Growing Degree Days (GDD) and Killing Degree Days (KDD) for selected spatiotemporal observations. Figure 5 displays these relationships for a single year (2009), while Figure 6 covers one state (MN) across all years in the sample (see appendix for more samples). Each point represents the paired values of a weather metric (e.g., GDD, KDD) calculated under the two approaches (pre- vs. post-interpolation). Points above the 45-degree line indicate that the post-interpolation metric exceeds its pre-interpolation counterpart, and vice versa.

The scatter plots show a clear pattern. Growing Degree Days (GDD) align closely around the 1:1 line, implying relatively small discrepancies between pre- and post-interpolation values. In most cases, GDD differences remain modest. Killing Degree Days (KDD), however, are systematically higher in the pre-interpolation approach. In other words, majority of the points lie below the 45-degree line, indicating that once the raw data are interpolated before the non-linear temperature transformation, fewer days exceed the threshold for harmful heat events.

From an agronomic perspective, these findings are consistent with the notion that spatial smoothing can dilute the intensity of local extremes (Haylock et al., 2008). Since KDD captures exceptionally hot days, any underestimation of daily temperature maxima will reduce the calculated KDD. This has direct implications for yield impact estimates because it may yield a muted view of the risks posed by extreme heat.

4.2 Regression Estimates

Table 2 presents the regression estimates using the pre-transformed (pre-interpolated) and post-transformed (post-interpolated) weather measures on corn yields.

Across both specifications, GDD retains a positive, statistically significant effect on yield, indicating that beneficial heat increases corn yields. While the difference in coefficients between the pre- and post-approaches is itself statistically significant, the magnitude of this difference remains

relatively small. GDD spans a broad temperature range and thus tends to be a relatively large cumulative value, making any interpolation-related discrepancies less pronounced.

By contrast, KDD consistently shows a strong negative influence on yields. This aligns with agronomic and economic research that identifies a sharp drop in yield once temperatures exceed critical thresholds (Schlenker & Roberts, 2009; Tack et al., 2018). Although the difference in KDD estimates between the two specifications is not statistically significant, its coefficient appears stronger than GDD's partly because KDD spans a smaller temperature range, making the per-unit effect on yields more pronounced.

CDD, which reflects potential cold stress, has a negative and significant coefficient in the pre-interpolation specification, whereas the post-interpolation estimate is smaller and statistically insignificant. However, the difference between these two estimates is not statistically significant. One plausible explanation is that localized cold snaps or suboptimal temperatures get blurred out when daily temperature values are first spatially averaged, reducing the apparent negative effect on yields. Since crop growth can be delayed if temperatures remain below critical thresholds (Annan & Schlenker, 2015), capturing these cold events with greater accuracy yields a more pronounced negative impact on yield in the pre-interpolation approach, even though the overall difference in estimates across the two methods is not statistically different.

Extreme wet conditions, as captured by the SPI index, shows a negative and significant effect in the pre-interpolation specification, indicating that intense localized rainfall can damage crops through waterlogging, disease pressure, or delayed planting and harvest. In the post-interpolation specification, this effect becomes insignificant, likely because spatial smoothing dampens these highly localized precipitation extremes, and the difference in estimates across the two approaches is itself statistically significant. Meanwhile, the coefficient on extreme drought is not significant in either specification.

The interaction KDD $\times$ Extreme Wet is positive and significant in the pre-interpolation model, indicating that extreme rainfall might offset part of the heat stress, either by cooling through evapotranspiration or by ensuring adequate soil moisture (Lobell et al., 2011). However, in the post-interpolation specification, the coefficient becomes close to zero and insignificant. This shift

might indicate that smoothing underestimates the co-occurrence of high heat and high precipitation, i.e., fewer "extremely wet and extremely hot" days are captured when weather data is interpolated first.

The results show that non-linear weather measures are sensitive to the sequence of interpolation. While broad patterns remain intact (beneficial GDD, harmful KDD, negative extreme precipitation effects), the magnitude and significance of these impacts can vary. From a policy perspective, underestimating extremes may produce an outlook on yield resilience that is not true, potentially downplaying the need for heat or flood adaptation measures.

4.3 Yield Prediction Differences

An important practical question is how yield predictions differ if the underlying yield-response coefficients remain the same, but the weather measures switch from pre- to post-interpolation (or vice versa). This exercise isolates measurement differences rather than structural changes in the estimated climate sensitivities. Figures 7 and 8 illustrate these yield differences ((Pre) − (Post)) over time and by state.

Predictions from post-interpolated weather data consistently produce higher yields relative to pre-interpolated data, suggesting that the model perceives less extreme heat/drought stress when the non-linear transformations are applied after smoothing. Over several decades, the gap appears to widen modestly, possibly reflecting stronger extreme events in later years. Certain states (e.g., South Dakota, Wisconsin, Nebraska) exhibit larger discrepancies. These states experience more frequent temperature or precipitation extremes, thus any smoothing exerts a proportionally bigger impact on the computed metrics. In some years, the difference in predicted yields can be a few percentage points—potentially translating into millions of dollars in value if scaled across large acreage (Burke & Emerick, 2016).

These findings echo the broader literature that calls attention to measurement error in climate variables: even modest deviations in the tails of temperature or precipitation distributions can drive sizable differences in predicted agricultural outcomes (Lobell, 2013). Consequently, choosing an interpolation strategy is not a trivial modeling choice. The timing of non-linear transformations

can substantially affect yield projections in regions prone to extremes.

4.4 Decomposing the Sources of Yield Differences

To pinpoint which weather variables drive the yield differences identified in the subsection above, a Shapley decomposition (Shapley, 1953) is conducted which partitions the total discrepancy between pre- and post-based yield predictions among the individual weather measures: GDD, KDD, CDD, SPI–Drought, and SPI–Wetness.

Table 3 shows that the overall difference in predicted yields (Post – Pre) averages +3.599 bushels per acre across the full sample, implying that post-interpolation predicts yields higher by this amount. Notably, Killing Degree Days (KDD) accounts for 93% of this difference (+3.345 bu/ac). Because KDD measures the intensity of hot days beyond a threshold, any smoothing prior to transformation substantially reduces the frequency and magnitude of hot days. The model, therefore, perceives less heat-induced damage, pushing yield estimates upward.

Growing Degree Days (GDD) also contributes a positive share (+0.41 bu/ac), ∼11.4% of the total. While GDD is less extreme by definition, minor underestimation or overestimation still modifies yield outcomes. SPI–Wetness exerts a slightly negative effect (5.24%), indicating that post-interpolation occasionally captures fewer (or differently located) wet extremes, nudging yields down in those situations.

The dominance of KDD is particularly relevant in a warming climate. Past research has shown that heat extremes are projected to increase in both frequency and severity (IPCC, 2023b), and even a few days above critical temperature thresholds can dramatically reduce yields (Schlenker & Roberts, 2009). If routine data processing smooths out these extremes, risk models and impact assessments may be systematically biased toward underestimating climate vulnerability. Thus, the Shapley analysis quantifies and confirms that extreme heat metrics are the principal reason why post-interpolation yields differ so notably from pre-interpolation yields under the same coefficient structure.

5 Conclusion

This study examines how the sequence of weather-data interpolation influences the measurement of extreme weather and its estimated effects on agricultural productivity. By comparing pre-interpolated (raw weather measures derived prior to spatial interpolation) with post-interpolated (extreme metrics derived after spatial interpolation), we find that the order of interpolation significantly affects the representation of localized extremes—and, in turn, the impacts observed in yield models.

In particular, growing degree days (GDD), which typically reflects beneficial heat accumulation rather than stress, and extreme wet conditions both show statistically significant differences between pre- and post-interpolated measures, indicating that post-interpolation smoothing can understate localized weather variations in both temperature and precipitation extremes. Although the estimated coefficients for killing degree days (KDD) do not differ significantly across the pre- and post-interpolation approaches, our counterfactual analysis shows that extreme heat remains a major driver of yield outcomes. Even modest variations in how many days exceed critical temperature thresholds can substantially shift predicted yields, regardless of whether the overall coefficient difference appears small. This highlights the importance of accurately identifying localized extremes: capturing a few additional high-heat days can meaningfully alter model predictions, reinforcing the need to preserve local weather extremes in agricultural impact assessments.

The findings highlight the importance of accounting for interpolation-induced measurement errors when analyzing the impact of extreme weather events on agricultural outcomes. Future work should explore alternative interpolation strategies and extend the analysis to additional outcome variables and regions to facilitate a broader understanding of how measurement error due to interpolation influences weather impacts on agriculture and related sectors.

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References

Figures

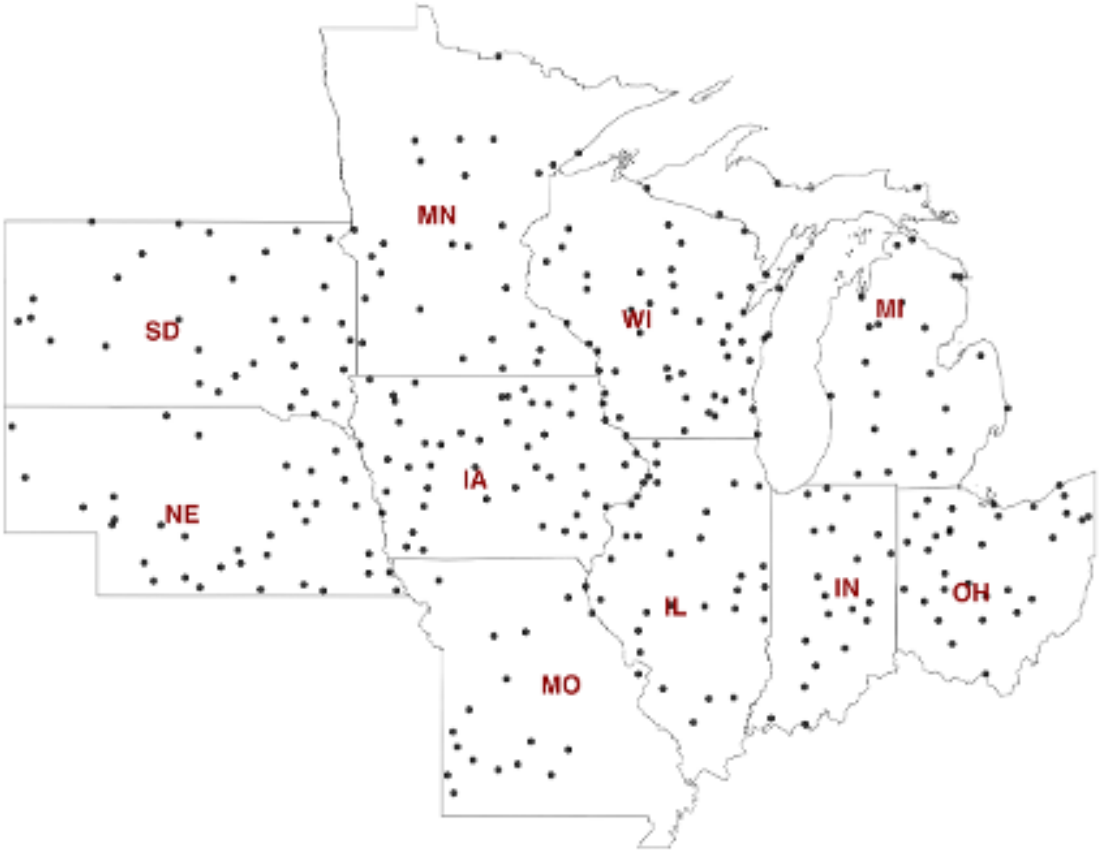
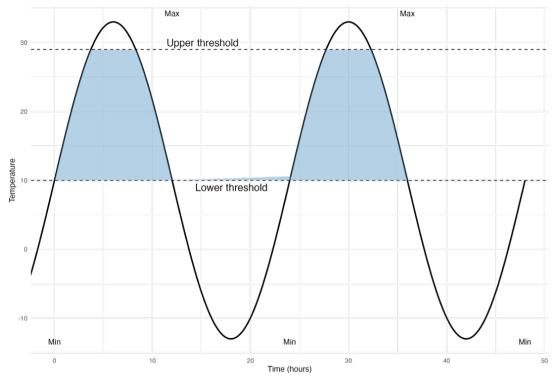
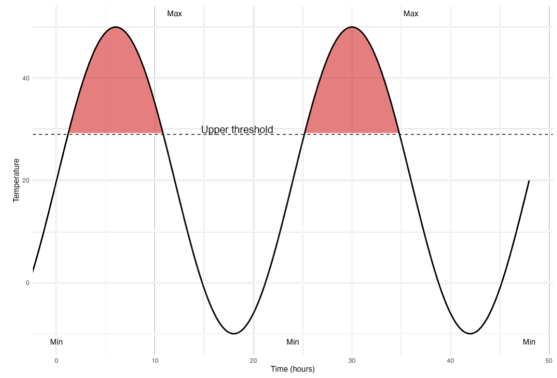


Figure 1: Distribution of weather stations across the Corn Belt Region (NOAA)



(a) Calculation of Growing Degree Days (GDD) using a sinusoidal function. The shaded area represents GDD for a two-day period using 10°C and 29°C as the lower and upper thresholds respectively.



(b) Calculation of Killing Degree Days (KDD) using a sinusoidal function. The shaded area represents KDD for a two-day period with 29°C as the threshold.

Figure 2: Calculation of Degree Days

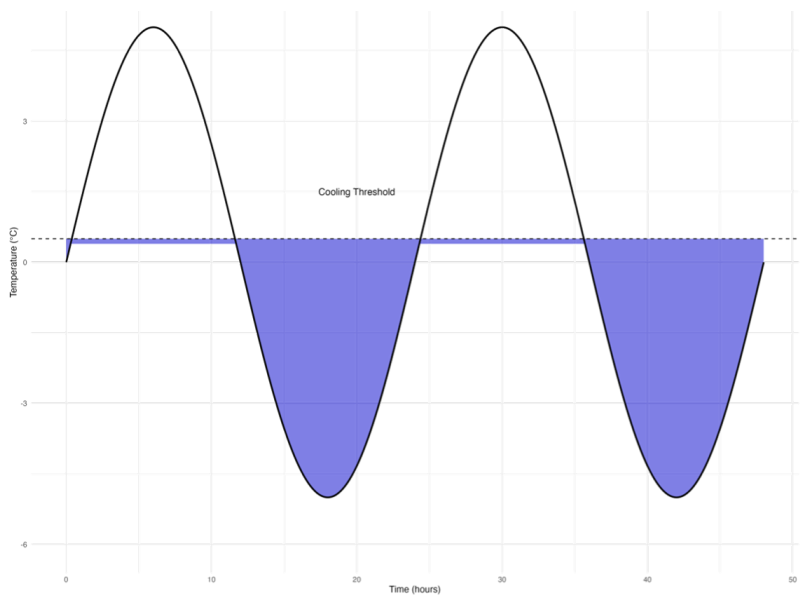


Figure 3: Calculation of Cooling Degree Days (CDD) using a sinusoidal function. The shaded area represents CDD for a two-day period with 0.5°C as the threshold.

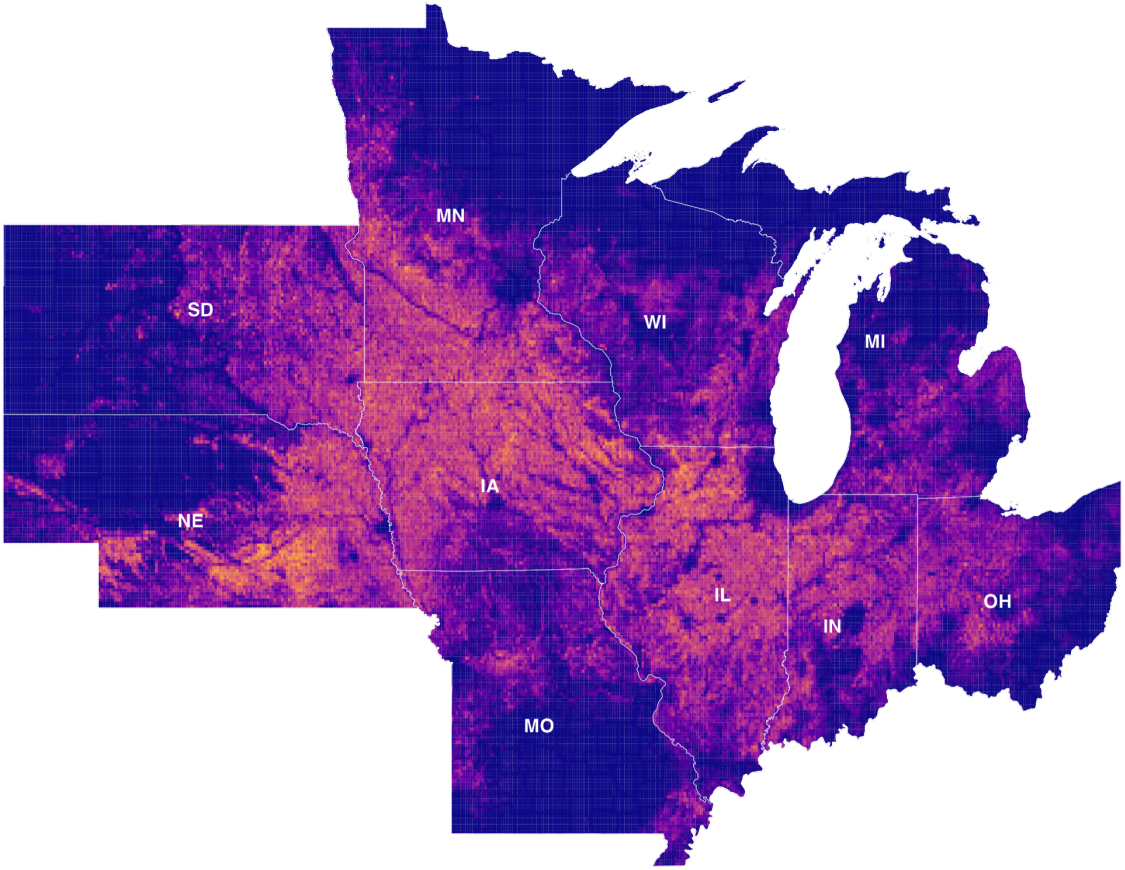
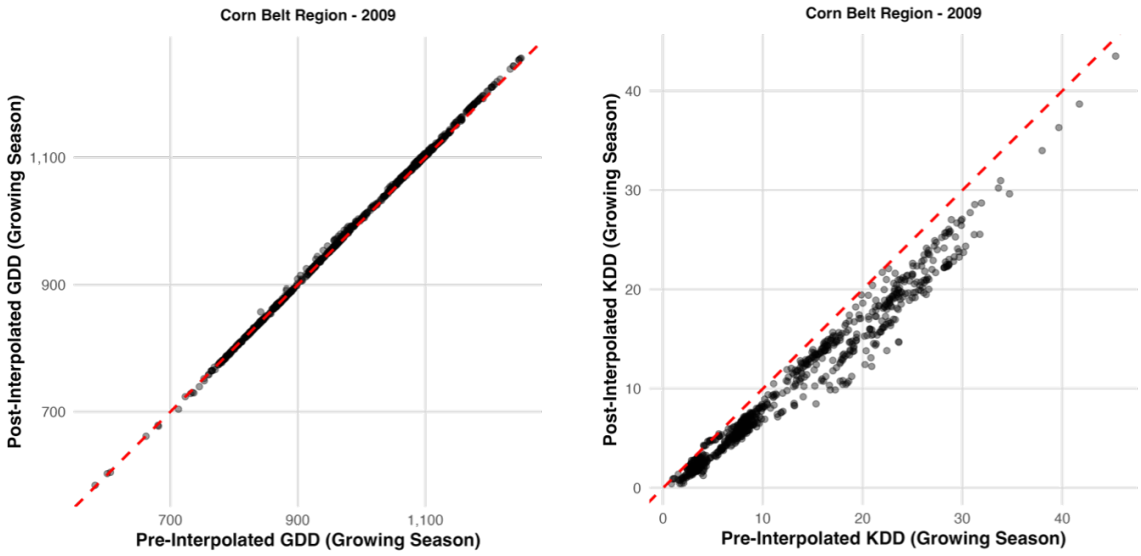
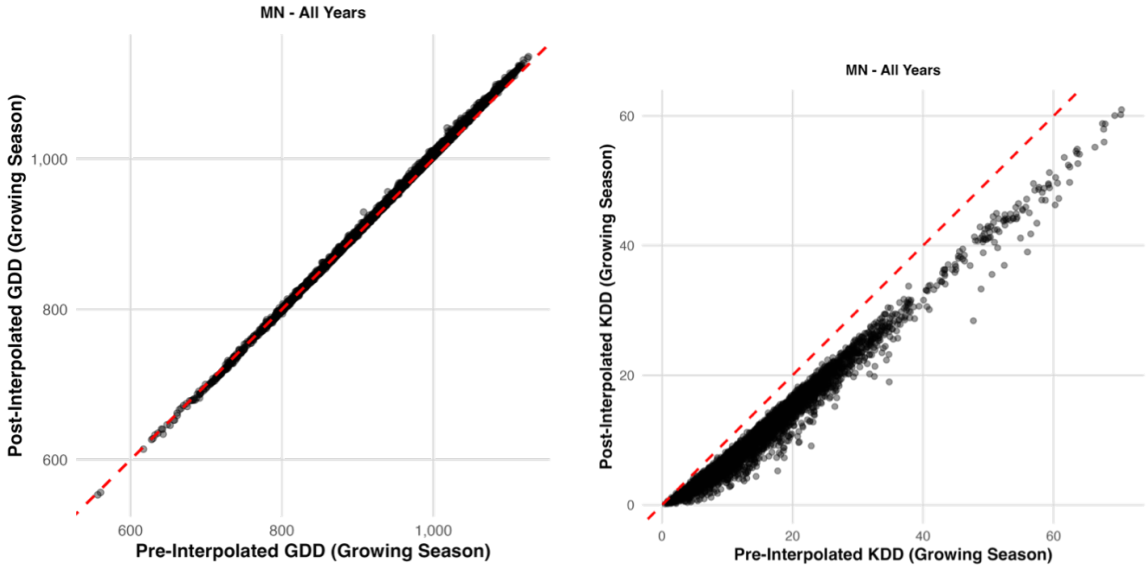


Figure 4: Corn share in the Corn Belt Region using Cropland Data Layer for the year 2023 and 0.05 x 0.05 grid resolution



(a) GDD (b) KDD

Figure 5: Scatter plots of (pre, post) pairs for GDD and KDD for sample year (2009) – Entire Region



(a) GDD (b) KDD

Figure 6: Scatter plots of (pre, post) pairs for GDD and KDD for sample state - Minnesota

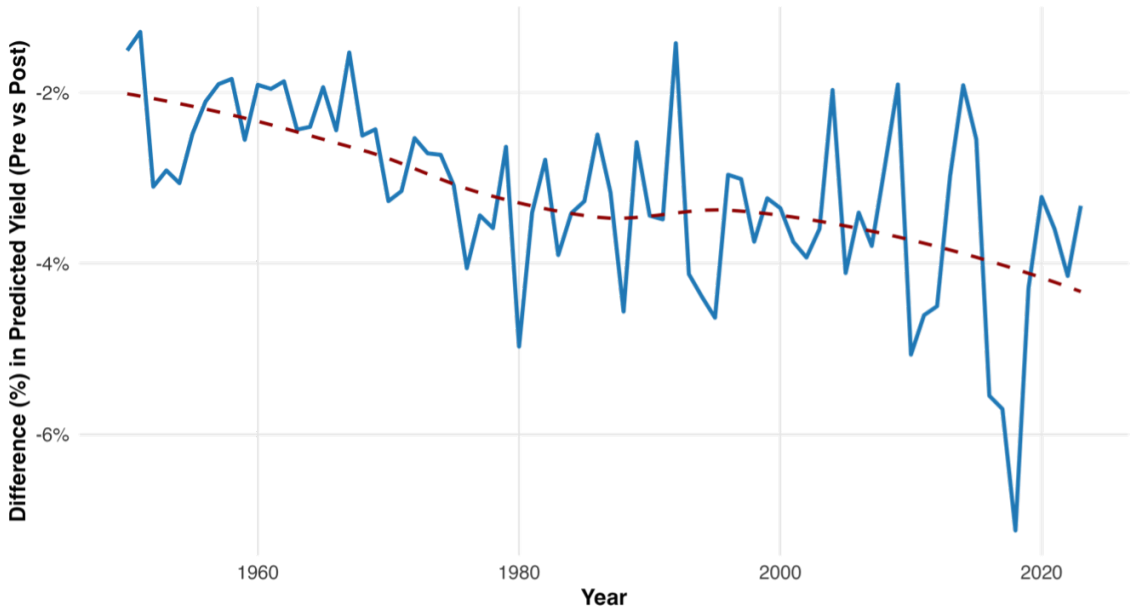


Figure 7: Difference (%) in predicted yield across time (pre vs post)

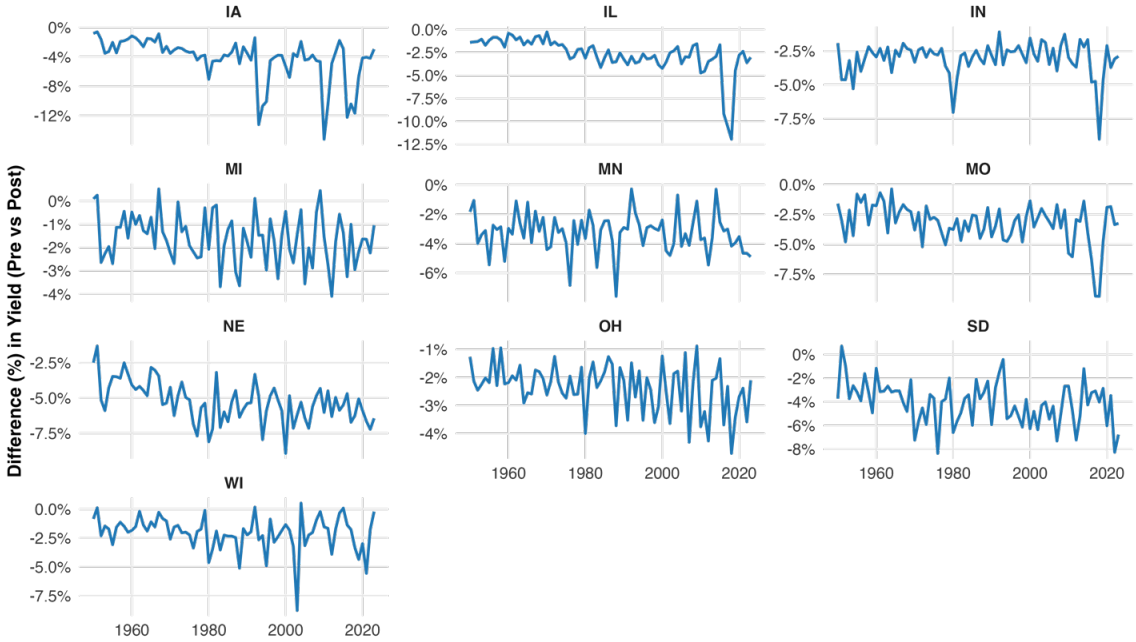


Figure 8: Difference (%) in predicted yield by state and time (pre - post)

Tables

Table 1: Classification of precipitation extremes based on the SPI index (Zhen et al., 2018)

SPI value	Category
$(2.0, +\infty)$	Extreme wet
$(1.5, 2.0]$	Severe wet
$(1.0, 1.5]$	Moderate wet
$(-1.0, 1.0]$	Normal
$(-1.5, -1.0]$	Moderate drought
$(-2.0, -1.5]$	Severe drought
$(-\infty, -2.0]$	Extreme drought

Table 2: Regression Results for Pre and Post Transformed Measures using Inverse Distance Weighting (IDW) Interpolation for Growing Season (1950-2023) – Growing Period

	Dependent Variable: log(yield)		
Variable	Pre-Transformed	Post-Transformed	Stat. Sig Difference
Growing Degree Days (GDD)	0.0007766**	0.0007296*	***
	(0.0002855)	(0.0002881)	
Killing Degree Days (KDD)	-0.0085963***	-0.0085658***	
	(0.0007119)	(0.0007586)	
Cooling Degree Days (CDD)	-0.0541041*	-0.0463203	
	(0.0226630)	(0.0721239)	
SPI (Extreme Drought)	0.1190238	0.0305934	***
	(0.1068375)	(0.0299817)	
SPI (Extreme Wet)	-0.3695180***	-0.0075054	***
	(0.0917226)	(0.0355996)	
KDD x Extreme Drought	-0.0014289	-0.0002336	*
	(0.0016508)	(0.0010737)	
KDD x Extreme Wet	0.0066186**	0.0000734	***
	(0.0019522)	(0.0010205)	
CDD x Extreme Drought	0.0149227	0.3414590	
	(0.0912078)	(0.5045463)	
CDD x Extreme Wet	-4.9351357	0.4322834	
	(4.9773041)	(0.2204557)	
Observations	61,499	61,499	
R2	0.868	0.868	
R2 Adjusted	0.866	0.866	
Clustered Standard Errors	by: county & year	by: county & year	
FE: County	YES	YES	
FE: Year	YES	YES	

*p ¡ 0.05, ** p ¡ 0.01, *** p ¡ 0.001

Table 3: Shapley Decomposition of the Average Predicted Yield Difference (Post – Pre)

Variable	Shapley Value (bu/ac)	% of Total Difference
KDD	+3.345	93.0%
GDD	+0.410	11.38%
CDD	+0.022	0.61%
SPI – Drought	+0.011	0.30%
SPI – Wetness	-0.189	-5.24%
Total Difference	+3.599	100%

A Appendix

Table 4: Seasons by state

State	Planting Begins	Planting Ends	Growing Ends	Harvest Begins	Harvesting Ends
IOWA	April 9th	May 21st	September 17th	September 17th	November 26th
ILLINOIS	April 2nd	June 2nd	September 10th	September 10th	November 26th
INDIANA	April 9th	June 11th	September 10th	September 10th	November 26th
MISSOURI	April 2nd	June 2nd	September 3rd	September 3rd	November 26th
MINNESOTA	April 9th	June 2nd	September 17th	September 17th	November 26th
S. DAKOTA	April 16th	June 2nd	September 17th	September 17th	November 26th
NEBRASKA	April 1st	June 1st	September 10th	September 10th	November 26th
OHIO	April 9th	June 11th	September 17th	September 17th	November 26th
WISCONSIN	April 16th	June 11th	September 17th	September 17th	November 26th
MICHIGAN	April 16th	June 11th	September 17th	September 17th	November 26th

This paper uses state-specific planting, growing, and harvesting periods instead of a single generalized season (see Table 4). The table shows seasonal timelines for planting, growing, and harvesting corn across the study region. The thresholds are taken from USDA-NASS such that the planting season is assumed to begin when more than 1% of the area in the state is shown to have planted corn and ends when over 90% of the area is shown to have planted corn. For instance, in Iowa, planting starts on April 9th and ends by May 21st, while in Illinois it begins earlier on April 2nd, continuing until June 2nd. The growing season follows, starting from the point when the planting reaches over 90% completion and extends to the start of harvesting. In Iowa, this period is from May 26th to September 17th, reflecting the primary growing months when corn matures. Finally, the harvesting season begins once over 1% of the area is harvested and continues until the end of the harvest, when over 90% is harvested. For example, if an extreme heat event occurs in early September, it may have a severe impact on states still in the peak growing season, like Minnesota or South Dakota, but less effect on states already entering their harvesting phase, like Missouri.

Table 5: Regression Results for Pre and Post Transformed Measures using Thin Plate Spline (TPS) Interpolation for Growing Season (1950-2023) – Growing Period

	Dependent Variable: log(yield)		
Variable	Pre-Transformed	Post-Transformed	Stat. Sig Difference
Growing Degree Days (GDD)	0.0008244**	0.0007885**	***
	(0.0002737)	(0.0002774)	
Killing Degree Days (KDD)	-0.0082767***	-0.0082558***	
	(0.0007745)	(0.0007632)	
Cooling Degree Days (CDD)	-0.0324743***	-0.0527395	
	(0.0076363)	(0.0452990)	
SPI (Extreme Drought)	0.1308851	0.0102737	***
	(0.0783545)	(0.0436586)	
SPI (Extreme Wet)	-0.2490377***	-0.0384090	***
	(0.0578185)	(0.0269266)	
KDD x Extreme Drought	-0.0006960	0.0000069	**
	(0.0011471)	(0.0013911)	
KDD x Extreme Wet	0.0045794***	0.0007178	***
	(0.0009653)	(0.0006918)	
CDD x Extreme Drought	0.1284521	-0.4962895*	
	(0.1754408)	(0.2317854)	
CDD x Extreme Wet	0.7274604***	0.1447661***	
	(0.1082026)	(0.0368987)	
Observations	61,505	61,505	
R2	0.868	0.867	
R2 Adjusted	0.866	0.865	
Clustered Standard Errors	by: county & year	by: county & year	
FE: County	YES	YES	
FE: Year	YES	YES	

*p ¡ 0.05, ** p ¡ 0.01, *** p ¡ 0.001

In addition to the IDW-based analyses, Table 5 presents regression results using Thin Plate Spline (TPS) interpolation for the growing season (1950–2023). Overall, the patterns are consistent with the IDW specification: GDD retains a positive and significant effect on corn yields; KDD shows a strong negative impact. Moreover, pre-transformed measures again tend to show greater sensitivity to extremes, reflected in the statistically different, larger or more significant coefficients for some of the key variables (e.g., SPI extreme wet). These TPS results support the broader conclusion that how and when weather data are interpolated can meaningfully influence the estimated effects of extreme weather on agricultural yields.

Table 6: (Irrigated vs Non-Irrigated) - Regression Results for Pre and Post Interpolation Models with County and Year Fixed Effects (IDW)

	Pre (Irrigated)	Pre (Non-Irrigated)	Post (Irrigated)	Post (Non-Irrigated)
Growing Degree Days	0.000361	0.000088	0.000551	0.000262
	(0.000639)	(0.000712)	(0.000666)	(0.000768)
Killing Degree Days	-0.005939***	-0.005913**	-0.005691**	-0.005833*
	(0.001532)	(0.001760)	(0.001857)	(0.002283)
Cooling Degree Days	-0.301332***	-0.256119***	-0.200044	-0.195760**
	(0.062389)	(0.067144)	(0.111761)	(0.065567)
SPI (Drought)	0.043589	0.018234	0.214683*	0.222594*
	(0.064253)	(0.067364)	(0.093509)	(0.100671)
SPI (Wetness)	-0.170891	-0.257566	0.189407*	0.139145
	(0.159084)	(0.145775)	(0.081994)	(0.083809)
KDD x Drought	-0.002145*	-0.002091*	-0.002364	-0.002496
	(0.000886)	(0.000906)	(0.001755)	(0.001897)
KDD x Wetness	0.005860*	0.008216**	-0.001966	-0.001350
	(0.002832)	(0.002580)	(0.001444)	(0.001517)
CDD x Drought	60.494542**	31.089704*	-0.895363*	-0.879364
	(18.028957)	(12.330292)	(0.351224)	(0.441196)
Observations	5,872	5,829	5,872	5,829
R2	0.896	0.901	0.895	0.899
R2 Adjusted	0.893	0.898	0.892	0.896
Std. Errors	County & Year	County & Year	County & Year	County & Year
FE: GEOID	YES	YES	YES	YES
FE: year	YES	YES	YES	YES

*p ¡ 0.05, ** p ¡ 0.01, *** p ¡ 0.001

As a further robustness check, Table 6 presents separate regressions for irrigated and non-irrigated cropland under both pre- and post-interpolation models. While the broad patterns remain consistent with the main analysis, particularly with the strong negative effect of KDD, several differences emerge. Notably, cooling degree days (CDD) shows stronger negative coefficients under irrigated conditions when measured in the pre-interpolated data, whereas post-interpolation results tend to reduce the magnitude or significance of certain extremes. Similarly, the SPI interactions vary across irrigation status, indicating that the ability to mitigate drought or wetness may depend on whether a field is irrigated. Overall, these findings reinforce that local weather extremes can be differentially captured once the data are interpolated, particularly for conditions like extreme heat or precipitation, and that irrigation status can moderate these effects.

Table 7: J-test Results Comparing Pre and Post Interpolation Models.

Model	Cross-model prediction	Coefficient	p-value	Significance
Pre-interpolation	Predicted yield from post	0.557 (0.252)	0.030	Yes
Post-interpolation	Predicted yield from pre	2.228 (0.573)	¡0.001	Yes

Note: Standard errors (SE) are in parentheses and clustered by county and year. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Predicted yield terms represent fitted values obtained from the alternative model, following Davidson-MacKinnon's J-test procedure for comparing non-nested models. We apply the Davidson and MacKinnon J-test (Greene, 1997; Davidson & MacKinnon, 1981) to compare the two models (pre vs post) for explaining county-level corn yield variations. The J-test assesses whether either set of variables provides additional explanatory power beyond what is captured by the other model. To perform the test, we first estimate each model independently, including fixed effects for counties and years. We then augment each model with predictions from the alternative model and assess whether these predicted values significantly improve explanatory power. The results, presented in Table 7, suggest that both pre- and post-interpolation measures contain unique explanatory power for yield variation.

Table 8: Tests for Equality of Coefficients (SUR-based Linear Hypothesis Tests)

Variable	F-statistic	p-value	Interpretation
GDD***	965.77	0.00000	Highly significant difference
KDD***	418.15	0.00000	Highly significant difference
CDD	0.1274	0.7211	No significant difference
SPI Drought***	110.76	0.00000	Highly significant difference
SPI Wetness***	26.899	0.00000	Highly significant difference

Note: *** indicates significance at the 1% level. Formal hypothesis tests based on Seemingly Unrelated Regressions (SUR) show statistically significant differences between pre- and post-interpolation model coefficients for Growing Degree Days (GDD), Killing Degree Days (KDD), and Standardized Precipitation Index (SPI) drought and wetness measures (Table 8). The differences in coefficients reveal that spatial interpolation can significantly change how these weather variables relate to yield variability, potentially through smoothing extremes or spatial heterogeneity in weather impacts.

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